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Enhanced Permeability Prediction Using Machine Learning Comparative Analysis and Uncertainty Quantification

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Abstract

This study aims to enhance permeability prediction in reservoir characterization by leveraging machine learning (ML) techniques, including Domain Transfer Analysis(DTA), Neural Networks (NN), and Multiple Linear Regression (MLR). The research integrates core data calibration with advanced feature selection methods to optimize predictive modeling. The objective is to compare these ML models in capturing permeability dynamics, quantify uncertainty propagation, and establish a framework for managing predictive errors, ultimately improving subsurface modeling accuracy for exploration and production applications.

A systematic workflow incorporates Ensemble Empirical Feature Selection (EEFS), Pearson's correlation analysis, and Backward Feature Elimination for optimal feature selection. The study integrates core data for model calibration and validation, ensuring reliability and practical relevance. Comparative analysis between Decision Tree, Neural Networks, and Multiple Linear Regression models evaluates their effectiveness in permeability prediction. Uncertainty quantification techniques are employed to assess predictive reliability and model sensitivity, providing deeper insights into permeability variation and uncertainty management across different geological settings.

Findings reveal that Neural Networks shows scatter in high and low permeability zones indicates reduced reliability in extrapolated regions, highlighting the NN model's sensitivity to input variability and the need for careful validation in unseen scenarios, while Decision Trees provides strong interpretability in understanding key permeability influences. Multiple Linear Regression exhibits limitations in handling complex permeability relationships, highlighting the need for feature selection optimization. The integration of core data significantly improves model accuracy, reducing bias in permeability predictions. Uncertainty analysis demonstrates varying sensitivity levels in ML models, allowing for better predictive risk management in reservoir characterization. The study offers a comprehensive framework for refining permeability models, supporting more informed decision-making in exploration and production strategies.

This research presents a unique integration of machine learning techniques with core data calibration, advancing permeability prediction beyond conventional models. The study introduces a systematic approach to uncertainty quantification, improving predictive accuracy and reliability. By evaluating ML model

limitations and strengths, this paper contributes to enhanced reservoir characterization methodologies, helping optimize subsurface characterization and modelling.

Introduction

Traditional approaches to permeability prediction in reservoir characterization, such as empirical correlations and linear regression models, often fall short in capturing complex, nonlinear relationships inherent in subsurface formations. These methods typically rely on simplified assumptions and are heavily dependent on deterministic parameters, which limit their ability to generalize across diverse lithologies and geological settings. Moreover, they frequently lack mechanisms for uncertainty quantification, resulting in biased predictions and reduced reliability at the field scale.

To address these limitations, this study explores the application of machine learning (ML) as a transformative approach for permeability estimation. ML models offer the capability to learn from large, multi-dimensional datasets and uncover hidden patterns that traditional methods may overlook. ML techniques can enhance prediction accuracy and provide deeper insights into reservoir behavior.

This research leverages the Microsoft Excel Co-Pilot Advanced Analysis module (Microsoft 2025) to evaluate a suite of statistical and ML models, including Domain Transfer Analysis(DTA), Neural Networks (NN), Multiple Linear Regression (MLR), Random Forest (RF), Support Vector Regression (SVR), K-Nearest Neighbors (KNN), and Gradient Boosting. Three models (DTA, NN, and MLR) were selected for in-depth analysis based on their alignment with field data. The study also incorporates uncertainty quantification to assess model sensitivity and predictive reliability. This integrated framework aims to refine permeability prediction workflows and support more informed decision-making in exploration and production strategies. The novelty of this study lies in its systematic comparison of these models using actual core data, combined with advanced feature selection techniques; Experience Eye Feature Selection (EEFS, [Arkalgud et al. 2021](#)), Pearson's Correlation (PC), and Backward Feature Elimination (BFE) to identify the most influential input variables.

Literature Review - ML in Petrophysics and Reservoir Modeling

Recent advancements in machine learning (ML) have significantly improved the accuracy and efficiency of petrophysical analysis and reservoir modeling. Traditional methods, such as empirical por-perm transforms and deterministic regression often fail to capture the nonlinear relationships and heterogeneity present in subsurface formations. To address these limitations, various ML algorithms have been applied to permeability prediction using well-log and core data.

[Asimiea et al. \(2023\)](#) conducted a comparative study of six ML models, including decision trees, support vector machines, and neural networks, for permeability prediction using open hole logs (GR, RHOB, NPHI, and RT). Their findings showed that Random Forest achieved the lowest mean squared error, outperforming other models in predictive accuracy and they proposed a workflow which integrated core and log data using artificial neural networks (ANNs). Their approach emphasized depth normalization and facies-based grouping to improve permeability prediction in uncored wells, addressing the limitations of traditional poro-perm transforms. [Milad et al. \(2021\)](#) extended permeability modeling to wells lacking NMR data by applying ML techniques such as neural networks and multilinear regression. They introduced the Large Pores Index (LPI) derived from standard logs, which significantly improved permeability estimates and supported better reservoir management decisions. [Khassaf et al. \(2025\)](#) introduced physics-informed machine learning (PIML) by embedding domain-specific constraints into tree-based ensemble models (XGBoost, CatBoost, Random Forest). Their approach minimized prediction errors and aligned closely with core measurements in carbonate reservoirs.

Prediction using DTA, Neural Networks and MLR

Arkalgud et al. (2021) introduced the "Experienced Eye" methodology, derived from Domain Transfer Analysis (DTA), for automated feature selection. This approach ranks input variables based on physical relevance rather than statistical averages, improving model precision and reducing noise. DTA was shown to outperform conventional statistical methods in feature selection and permeability modeling.

Neural Networks have been widely adopted for subsurface prediction. Eriavbe et.al. (2019) used ANNs to train models on core-calibrated data, demonstrating improved accuracy over traditional transforms. Milad et.al. (2021) further validated NN performance in complex carbonate reservoirs without NMR data, showing strong alignment with core permeability.

Multiple Linear Regression (MLR) has served as a baseline model in several studies (e.g., Asimiea et.al. 2023, Milad et.al. 2021). While computationally efficient, MLR consistently showed limitations in capturing nonlinear relationships and complex reservoir behavior.

What is Feature Selection?

Feature selection is a critical step in machine learning workflows, particularly in subsurface modeling, where datasets often contain numerous logging curves and derived parameters. It involves identifying the most relevant input variables (features) that contribute significantly to the prediction of a target variable, such as permeability. By reducing dimensionality, feature selection improves model accuracy, reduces computational cost and mitigates the impact of irrelevant or noisy data (Arkalgud et al., 2021).

This study employed three feature selection techniques to optimize permeability prediction:

1. **Experienced Eye Feature Selection (EEFS)**
2. **Pearson's Correlation (PC)**
3. **Backward Feature Elimination (BFE)**

EEFS is a physics-informed method derived from Domain Transfer Analysis (DTA). Unlike statistical approaches, EEFS ranks input features based on their physical relevance and influence on the target variable. It maintains the discrete identity of each input and evaluates its contribution through a parametric solution space. EEFS has been shown to outperform conventional methods by reducing noise and improving model precision (Arkalgud et al., 2021).

Pearson's correlation (PC) measures the linear relationship between each input feature and the target variable. It provides a quick statistical assessment of feature importance, with values ranging from -1 (strong negative correlation) to +1 (strong positive correlation). While effective for identifying linear dependencies, it may overlook nonlinear relationships and is sensitive to outliers (Arkalgud et al., 2021).

BFE is a wrapper method that starts with all available features and iteratively removes the least significant ones based on statistical metrics such as p-values or R^2 . This method is computationally intensive but can yield highly optimized feature subsets tailored to the specific prediction model (Arkalgud et al., 2021).

Justification for Method Selection

These three methods were chosen to balance physical relevance, statistical robustness, and computational efficiency:

- EEFS was selected for its ability to incorporate domain knowledge and resolve feature influence in a physically consistent manner.
- Pearson's Correlation provided a fast and interpretable baseline for feature ranking.
- BFE offered a model-specific optimization strategy to refine input selection and improve predictive accuracy.

Data and Methodology

This study was conducted using a combination of open-hole log data and core measurements collected from three onshore fields of Abu Dhabi, UAE. A type-log is shown in Figure 1.

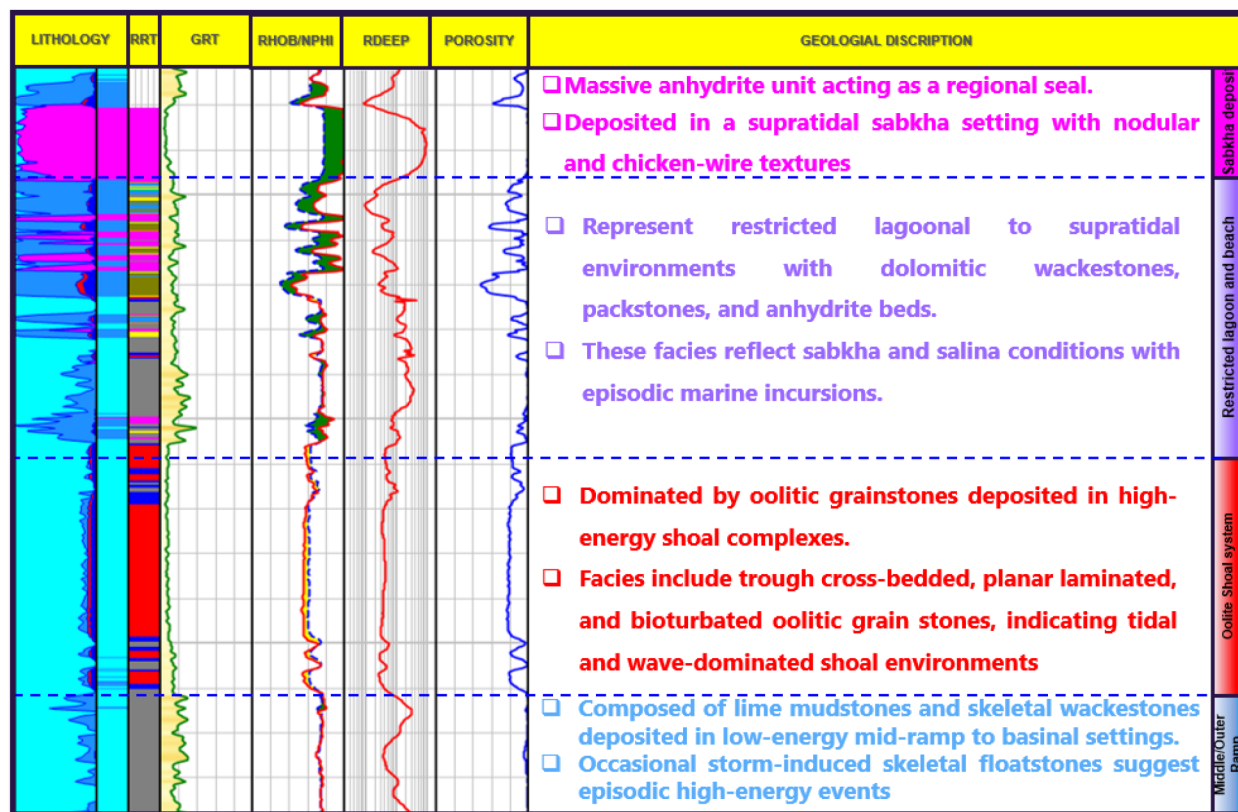


Figure 1—Type log for reservoir

The dataset represents different zones that represent variable depositional environments within the Upper Jurassic carbonate reservoir, including sabkha, restricted lagoon, beach deposits, oolitic shoal systems, and mid-to-outer ramp settings. The reservoir quality is significantly influenced by diagenetic processes such as dolomitization, cementation (calcite, saddle dolomite, anhydrite), and stylolitization. Although thermochemical sulfate reduction and hydrothermal fluid activity have been identified, their volumetric impact is minor (Hollis, et.al., 2017).

The primary objective of this study is to develop machine learning models capable of capturing the full spectrum of permeability uncertainty across the fields and aim to identify patterns and controlling factors affecting permeability by applying a range of statistical and predictive modeling techniques. This ensures that the final models are robust and capable of representing the inherent variability and uncertainty in permeability distribution.

Tools and Software

Interactive Petrophysics software was used to pre-process log and core data and to build MLR models for permeability modeling.

Statistical analysis was conducted using the Co-pilot Excel (Advanced Analysis) module. This included outlier detection via the IQR method, univariate distribution analysis, and multivariate trend identification. The exploratory data analysis (EDA) provided insights into relationships among different variables, prior to selecting MLR models for the input selection for training and prediction.

Data Processing

The dataset utilized in this study comprises well log and core from sixteen wells along with laboratory measurements including Routine Core Analysis (RCA), Mercury Injection Capillary Pressure (MICP), and thin petrographic section analysis. The preprocessing workflow was designed to ensure consistency, accuracy, and data readiness.

1. Well log data for 16 wells (GR, NPHI, RHOB, RT, RXO, PEF).
2. Core data for 16 wells
3. RCA of ~3500 RCA data points
4. MICP of 154 data points (used in developing SRT)
5. Facies and diagenesis data based on 253 thin sections.

Well Log Data QC and Conditioning

Gamma Ray (GR), Neutron Porosity (NPHI), Bulk Density (RHOB), Photoelectric (PEF), True Resistivity (RT) and Flushed Zone Resistivity (RXO) were subjected to quality control procedures. Spurious spikes, null values, and depth mismatches were corrected using interpolation and smoothing techniques.

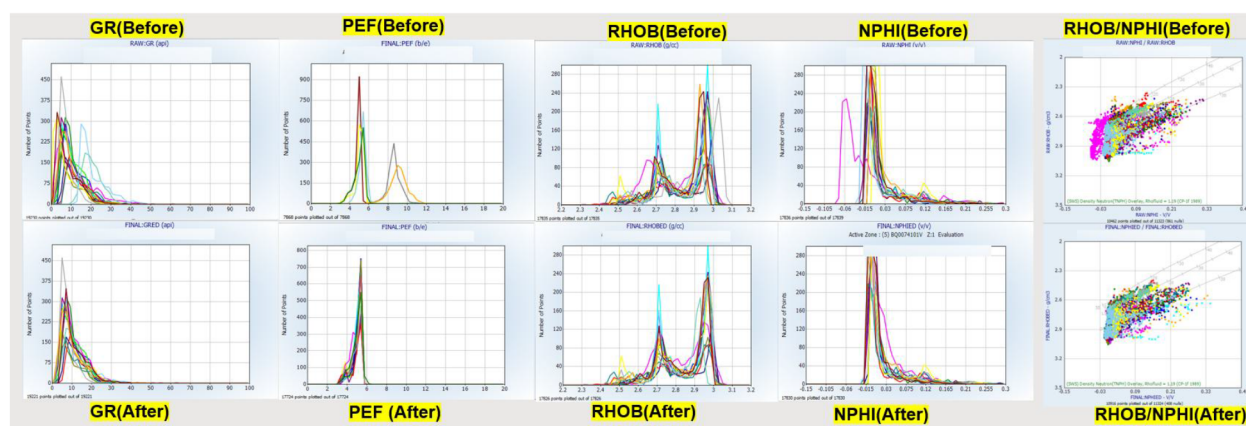


Figure 2—Well log normalization (GR, PEF, RHOB, NPHI)

Core Data QC and Conditioning

Routine Core Analysis (RCA) measurements from sixteen wells were evaluated using multiple cross-plots to identify and eliminate outliers that could compromise data integrity. From ~3500 initial data points, 1,885 were flagged as outliers based on statistical and visual QC methods. After thorough quality control, a filtered dataset of ~1100 reliable data points were retained.

This dataset (Figure 3) shows Field A and B data that was used in core-log data integration. Core data was depth-matched with corresponding log data to ensure alignment.

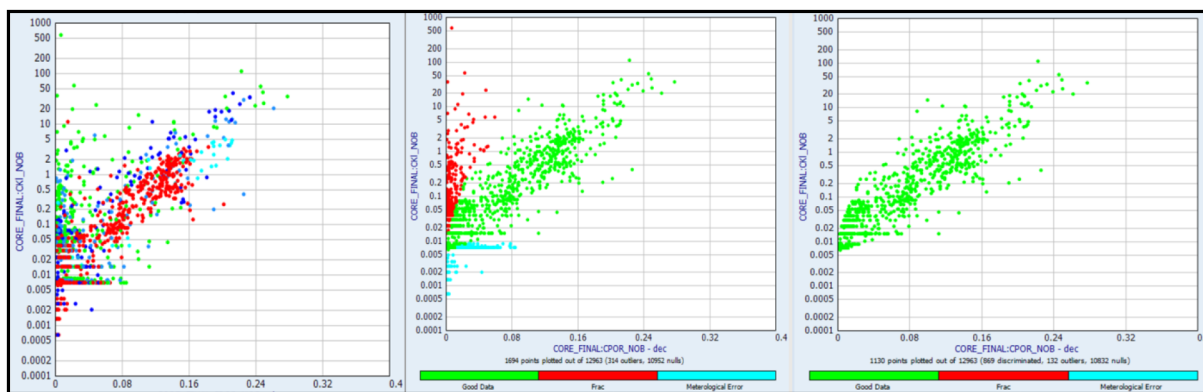


Figure 3—Core Porosity & Permeability a) all data by well b) RCA data color code by QC Flag c) RCA data finalized

Descriptive statistics

The log/core dataset was segmented into data frames; each linked to a variable library in Python to facilitate visualization of data trends. These data frames were grouped by LITHO, Static Rock Type (SRT - a classification derived from thin section and MICP data integration) and ZONE, which represents geological sequence boundaries indicative of varying depositional environments.

MICP Saturation Curves, Porosity-Permeability and PTR plots for LITHO-1 and LITHO-2 are color coded and shown in Figure 4.

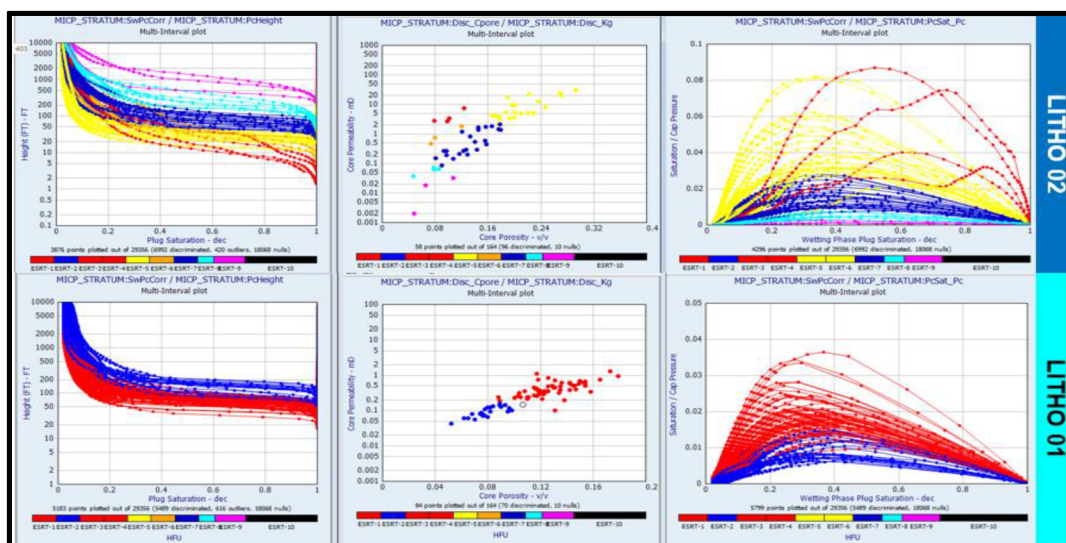


Figure 4—MICP Saturation Curves, Porosity-Permeability and PTR plots (LITHO-1, LITHO-2)

- LITHO-01 grain density ≤ 2.74 (i.e. limestone dominated)
- LITHO-02 ≥ 2.74 (higher probability of dolomite)

LITHO was used to define a data frame to perform exploration data analysis (EDA – see Appendix). Log and core statistics for LITHO-01 and LITHO-02 are shown in Table 1.

Table 1—Log and core statistics for a) LITHO-01 and b) LITHO-02

LITHO-01	CPOR	PEF	TVDSS	GR	RT	RHOB	NPHI	RXO	KL
mean	0.11	6.06	10952.09	6.35	57.62	2.60	0.08	13.27	0.43
std	0.03	1.61	153.01	3.43	220.70	0.04	0.04	49.09	1.84
min	0.00	4.50	10615.43	1.71	1.90	2.54	-0.01	1.11	0.01
25%	0.09	5.00	10917.67	4.21	9.46	2.57	0.06	2.32	0.11
50%	0.11	5.00	10967.30	5.17	14.09	2.59	0.08	3.53	0.24
75%	0.12	8.26	11010.50	7.60	51.96	2.61	0.11	5.29	0.39
max	0.17	9.80	11277.52	22.88	2664.74	2.74	0.18	438.47	30.46
LITHO-02	CPOR	PEF	TVDSS	GR	RT	RHOB	NPHI	RXO	KL
mean	0.12	4.52	10902.20	13.43	285.11	2.72	0.14	107.34	4.07
std	0.06	1.26	192.06	5.14	1529.17	0.09	0.10	409.46	10.61
min	0.00	2.89	10494.60	1.74	1.45	2.54	-0.03	0.26	0.02
25%	0.07	4.00	10768.76	10.02	8.60	2.66	0.06	1.42	0.10
50%	0.13	4.00	10880.40	12.73	27.92	2.70	0.12	4.42	0.62
75%	0.17	5.00	11007.23	15.77	156.16	2.76	0.22	16.79	3.16
max	0.29	8.19	11464.73	32.97	22056.13	2.99	0.35	4606.04	92.71

Log core statistics for split by zone are shown in Table 2.

Table 2—Log and core statistics for a) ZONE-02 / 03 and b) ZONE-05

ZONE 02-03	CPOR	PEF	TVDSS	GR	RT	RHOB	NPHI	RXO	KL
mean	0.12	4.67	10886.62	13.58	295.91	2.72	0.13	110.11	3.51
std	0.06	1.44	188.90	5.14	1507.18	0.09	0.10	401.71	8.81
min	0.00	2.89	10494.60	1.74	1.45	2.54	-0.03	0.26	0.01
25%	0.07	4.00	10766.38	10.09	9.14	2.66	0.05	1.52	0.09
50%	0.12	4.00	10867.67	12.85	27.90	2.70	0.11	5.02	0.53
75%	0.17	5.23	10989.14	16.15	157.24	2.76	0.21	20.41	3.00
max	0.29	9.80	11464.73	32.97	22056.13	2.99	0.35	4606.04	92.71
ZONE 05	CPOR	PEF	TVDSS	GR	RT	RHOB	NPHI	RXO	KL
mean	0.11	5.99	10966.38	5.98	40.57	2.59	0.08	7.68	0.74
std	0.03	1.58	148.35	2.76	109.48	0.03	0.04	24.60	5.43
min	0.00	4.00	10644.09	1.71	1.90	2.54	-0.01	1.11	0.02
25%	0.09	5.00	10932.99	4.16	9.45	2.57	0.06	2.31	0.13
50%	0.11	5.00	10973.70	5.06	13.47	2.59	0.08	3.36	0.25
75%	0.12	8.21	11013.73	7.30	51.74	2.60	0.11	4.76	0.40
max	0.21	9.11	11282.85	20.70	1602.40	2.72	0.20	318.46	79.00

- ZONE-02-03 was predominantly deposited in a restricted lagoon and beach
- ZONE-05 was deposited in an oolite shoal environment.

Exploration Data Analysis (EDA)

Detailed EDA was carried out on the dataset using the following techniques:

- Heatmaps - for correlation insights that help identify key numerical
- Histograms - for distribution patterns
- Box plots - for outlier detection and spread
- Cross plots - for variable relationships
- Statistical Models – to understand each model sensitivity to output variable

Exploratory Data Analysis (EDA) provides a foundational understanding of a dataset through various visualization techniques. Heatmaps are used to uncover correlations between numerical variables, helping identify strong relationships and potential multicollinearity that could affect modeling. Histograms reveal the distribution of individual features, highlighting skewness, central tendencies, and outliers. Box plots offer a compact view of data spread and variability, making it easier to detect extreme values and compare

distributions across categories. Lastly, cross plots (scatter plots) are instrumental in examining relationships between pairs of variables, revealing trends, clusters, and separability that support feature selection and model validation. Together, these techniques enable a comprehensive and intuitive understanding of the data, guiding further analysis and decision-making.

All analysis conducted under EDA has been defined with details in appendices.

Results and Discussion

Model Performance

The model was built using core data from three fields for the same reservoir to capture lateral and vertical variation for model building and to analyze the performance of the trained model on a regional scale.

Statistical Models (DTA, NN and MLR) were tested using feature selection methods combinations (EEFS, BFE, PC and User) using the top 6 curves that show the highest correlation for both training and test data with an aim to get $\pm 10\%$ deviation to validate the stability of the model.

Table 3 presents the R-squared values for models built using the top 3, 4, 5, and 6 input curves, comparing the performance of Domain Transfer Analysis(DTA), Multiple Linear Regression (MLR), and Neural Networks (NN) across both training and test datasets.

- **For training data:**

- The highest performing models are DTA ($R^2 > 0.82$)
- The poorest R^2 metrics are from MLR (R^2 of 0.62 to 0.66)
- The highest R^2 is 0.8854 when using the top 5 curves and DTA-BFE
- Using 6 curves does not materially improve the R^2 for the DTA techniques and in the case of DTA-PC and DTA-User, it reduces R^2 .

- **For test data**

- The highest performing models are DTA (R^2 of 0.73 – 0.79)
- The poorest R^2 metrics are from MLR (R^2 of 0.49 to 0.54)
- The highest R^2 is 0.7938 using the top 3 curves and DTA-EEFS
- Additional curves using DTA-EEFS reduces the R^2 metric (potential overfitting)

Table 3—R Square results for the top 3, 4, 5 and 6 curves using DTA / MLR and NN models (a) Training and (b) test data

Curves	DTA-EEFS	DTA-BFE	DTA-PC	DTA-User	MLR-EEFS	MLR-BFE	MLR-PC	MLR-User	NN-EEFS	NN-BFE	NN-PC	NN-User
Top 3 curves	0.8632	0.8498	0.8262	0.8262	0.6286	0.6286	0.6566	0.6499	0.7251	0.7518	0.7484	0.7664
Top 4 curves	0.8556	0.8613	0.8572	0.8593	0.6318	0.6318	0.6558	0.6494	0.7548	0.7293	0.7437	0.7551
Top 5 curves	0.8696	0.8854	0.8854	0.8826	0.6322	0.6355	0.6355	0.6355	0.7647	0.7238	0.7318	0.7342
Top 6 curves	0.8891	0.8806	0.8806	0.8814	0.6333	0.6333	0.6333	0.6333	0.7267	0.7707	0.7517	0.7660

Curves	DTA-EEFS	DTA-BFE	DTA-PC	DTA-User	MLR-EEFS	MLR-BFE	MLR-PC	MLR-User	NN-EEFS	NN-BFE	NN-PC	NN-User
Top 3 curves	0.7938	0.7859	0.7632	0.7323	0.5247	0.5247	0.4968	0.4876	0.6846	0.6982	0.7055	0.7032
Top 4 curves	0.7988	0.7879	0.7690	0.7555	0.5253	0.5253	0.5245	0.4877	0.7100	0.6981	0.7024	0.7042
Top 5 curves	0.7801	0.7566	0.7565	0.7572	0.5259	0.5363	0.5363	0.5159	0.7098	0.6736	0.6985	0.6940
Top 6 curves	0.7715	0.7648	0.7648	0.7648	0.5350	0.5350	0.5350	0.5350	0.6848	0.7168	0.6986	0.7135

Model performance was further evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), with the results summarized in [Table 4](#)

Table 4—Metrics for DTA / MLR and NN models (a) Mean Average Error (MAE) and (b) Root Mean Square Error (RMSE):

Curves	DTA-EEFS	DTA-BFE	DTA-PC	DTA-User	MLR-EEFS	MLR-BFE	MLR-PC	MLR-User	NN-EEFS	NN-BFE	NN-PC	NN-User
Top 3 curves	0.1096	0.1240	0.1476	0.1372	0.2461	0.2461	0.2088	0.2098	0.1895	0.1772	0.1765	0.1710
Top 4 curves	0.1242	0.1184	0.1187	0.1266	0.2436	0.2436	0.2089	0.2101	0.1776	0.1825	0.1779	0.1762
Top 5 curves	0.1184	0.0997	0.0997	0.1076	0.2435	0.2424	0.2424	0.2103	0.1756	0.1867	0.1823	0.1839
Top 6 curves	0.0963	0.1034	0.1034	0.1034	0.2428	0.2428	0.2428	0.2428	0.1883	0.1708	0.1794	0.1768
Curves	DTA-EEFS	DTA-BFE	DTA-PC	DTA-User	MLR-EEFS	MLR-BFE	MLR-PC	MLR-User	NN-EEFS	NN-BFE	NN-PC	NN-User
Top 3 curves	0.2766	0.2928	0.3216	0.3167	0.7963	0.7963	0.5471	0.5620	0.3959	0.3750	0.3659	0.3662
Top 4 curves	0.2882	0.2842	0.2853	0.2912	0.7797	0.7797	0.5853	0.5639	0.3723	0.3793	0.3691	0.3691
Top 5 curves	0.2789	0.2555	0.2555	0.2619	0.7784	0.7974	0.7974	0.6368	0.3739	0.3839	0.3789	0.3824
Top 6 curves	0.2483	0.2609	0.2609	0.2600	0.8057	0.8057	0.8057	0.8057	0.3854	0.3604	0.3697	0.3781

- **MAE**
 - Highest performing models are DTA (MAE from 0.09 to 0.15)
 - The poorest metrics are from MLR (MAE from 0.21 to 0.25)
- **RMSE**
 - Highest performing models are DTA (RMSE from 0.25 to 0.32)
 - The poorest metrics are from MLR (RMSE from 0.55 to 0.81)

The comparison of predicted permeability from DTA with Pearson's Correlation (DTA-PC) versus core data is shown in [Figure 5](#).

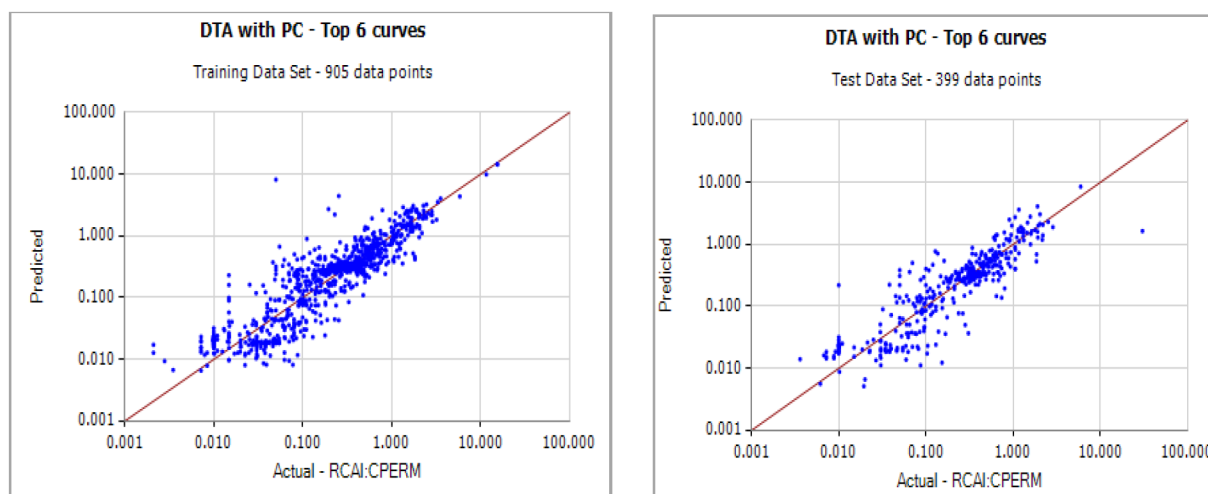


Figure 5—Permeability prediction – y axis = prediction, x-axis = core: a) Training data b) Test data

Machine Learning (ML) Experience Eye models

Trained models including Decision Tree Algorithm (DTA), Neural Networks (NN), and Multiple Linear Regression (MLR) were exported and used for permeability prediction that included feature selection.

To assess the generalization capability of these models, Field C core data were deliberately excluded during training and prediction phases at well level model ([Figure 6](#)). This blind testing approach enabled evaluation of the models' predictive performance on unseen data.

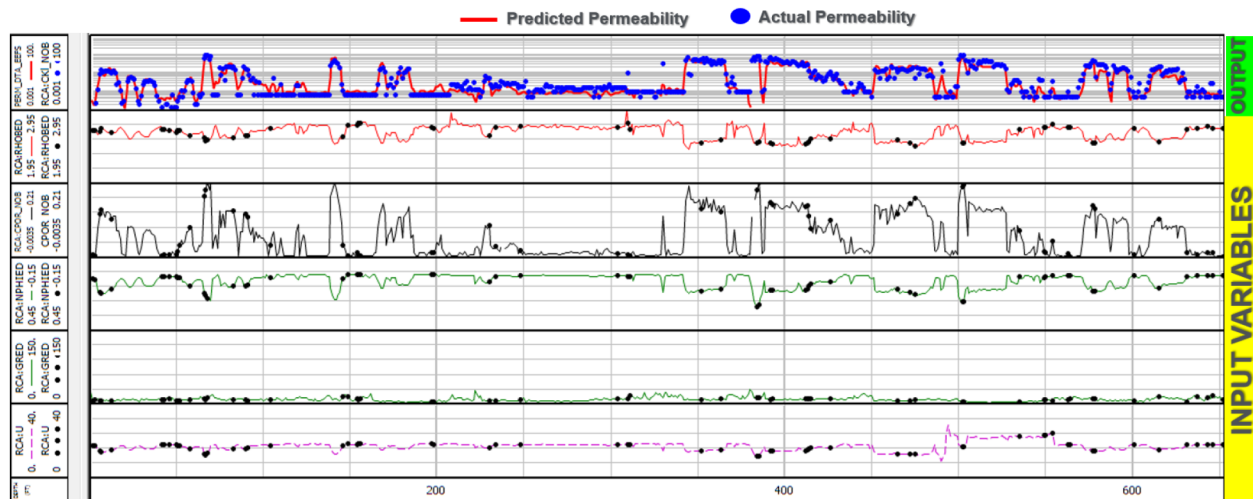


Figure 6—Trained Well from Field A & B – data sorted by TVDSS to captured perm trend with depth – black

Track 1 Depth

Track 2 $U = \text{PEF} \times (\text{RHOB} + 0.1883) \times 0.93423$

Track 3 Gamma Ray

Track 4 Neutron

Track 5 Porosity

Track 6 Bulk Density

Track 7 Core Permeability (blue point) & Predicted Permeability (red line)

Multiple permeability models were developed and benchmarked against actual data to assess the variability and statistical uncertainty inherent in each approach. This comparative analysis enabled the identification of the model that best conforms to a normal error distribution, based on systematic evaluation of prediction errors across all models.

- DTA 3 models: including EEFS, BFE, PC feature selection in prediction scenario
- NN 1 model: excluded EEFS, BFE and PC feature selection in prediction scenario
- MLR 1 model: excluded EEFS, BFE and PC feature selection in prediction scenario

Machine Learning Model Results

DTA Model Prediction results

Comparative analysis using multiple feature selection strategies confirmed the model's generalization capability across wells with blind core data (Figure 7). The predicted permeability values showed strong linear alignment with measured data, indicating high predictive accuracy. To quantify the uncertainty embedded in the model, envelope bounds were introduced using scaling factors of $Y \times 3$ (blue line) and $Y \times 0.33$ (green line). These bounds effectively captured the full range of prediction uncertainty, providing a robust framework for evaluating model reliability under blind test conditions.

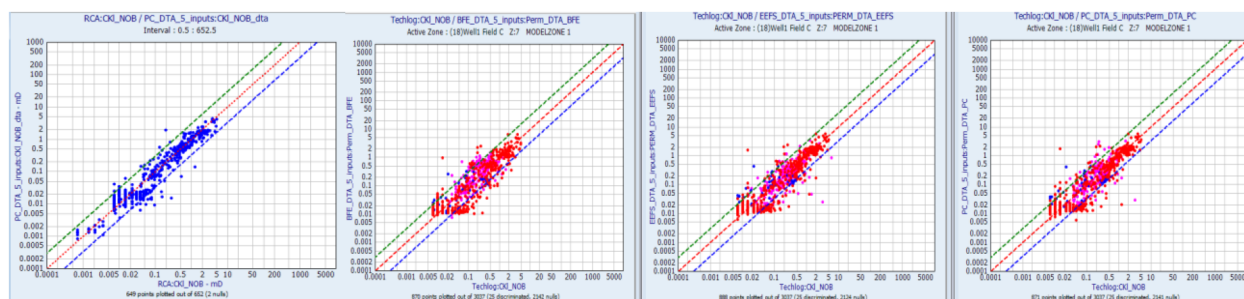


Figure 7—Permeability prediction – y axis = RCA, x-axis = prediction: a) Trained Well Field A & B b) DTA-BFE-color by Field c) DTA_EEFS-color code by Field d) DTA-PC code by Field – Color legend: Field A (Red), Field B (Blue) & Field C (Pink)

Neural Network Model Prediction results

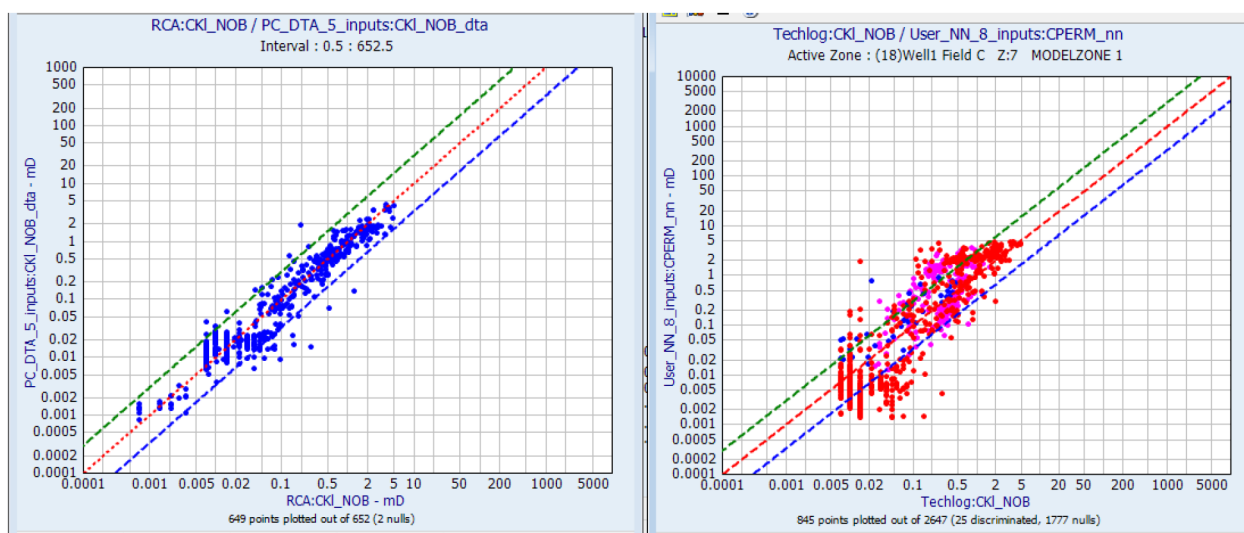


Figure 8—Permeability pred – y axis = RCA, x-axis = prediction: a) Trained Feild A & B Well b) NN Model with blind wells – Field A (Red), Field B (Blue) and Field C (Pink)

In general, the NN model captures permeability trends (Figure), however the scatter reveals noticeable deviations, particularly in zones of high and low permeability suggesting non-stabilization and reduced reliability in extrapolated regions. This highlights the NN model's sensitivity to input variability.

Multi Linear Model Prediction results

The Multiple Linear Regression (MLR) model, when tested using the Experience Eye Feature Selection (EEFS), yielded lower R^2 values, indicating poor alignment with the training data. To investigate this further, individual MLR models were built outside the EEFS framework, which showed significantly improved R^2 metrics (Figure 9).

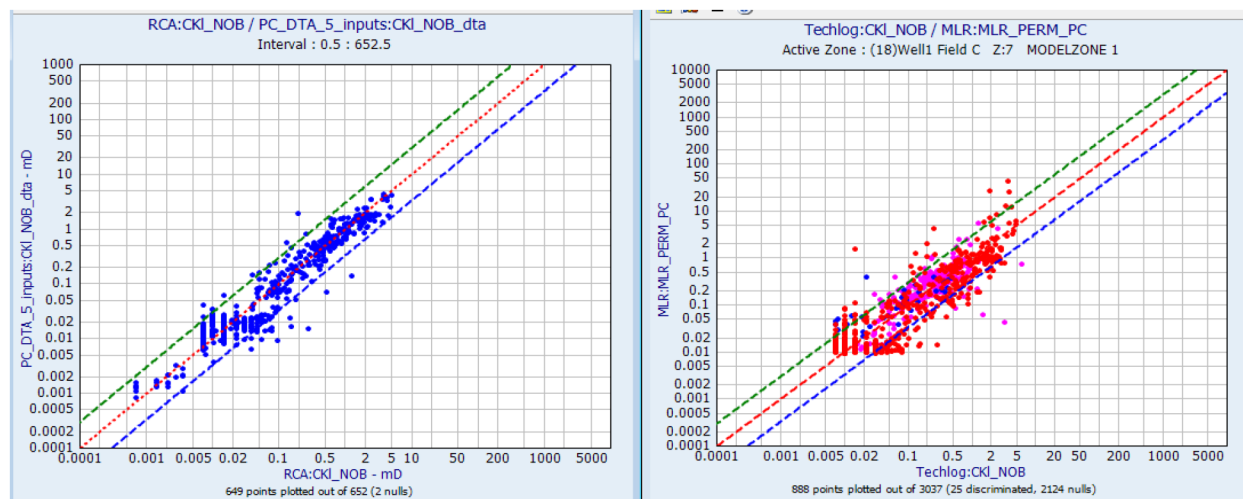


Figure 9—Permeability pred – y axis = RCA, x-axis = prediction: a) Trained Feid A & B Well b) NN Model with blind wells

This contrast suggests that linking MLR models through EEFS introduces higher uncertainty, likely due to the linear nature of MLR being incompatible with the physics-informed ranking of features in EEFS. Therefore, while MLR may perform adequately in isolated scenarios, its integration within the EEFS-driven workflow may not be optimal and should be approached with caution.

Comparison plots of model results and core data, for wells from Field A and Field B which were used for prediction analysis using trained well Field A & B, is shown in Figure 10.

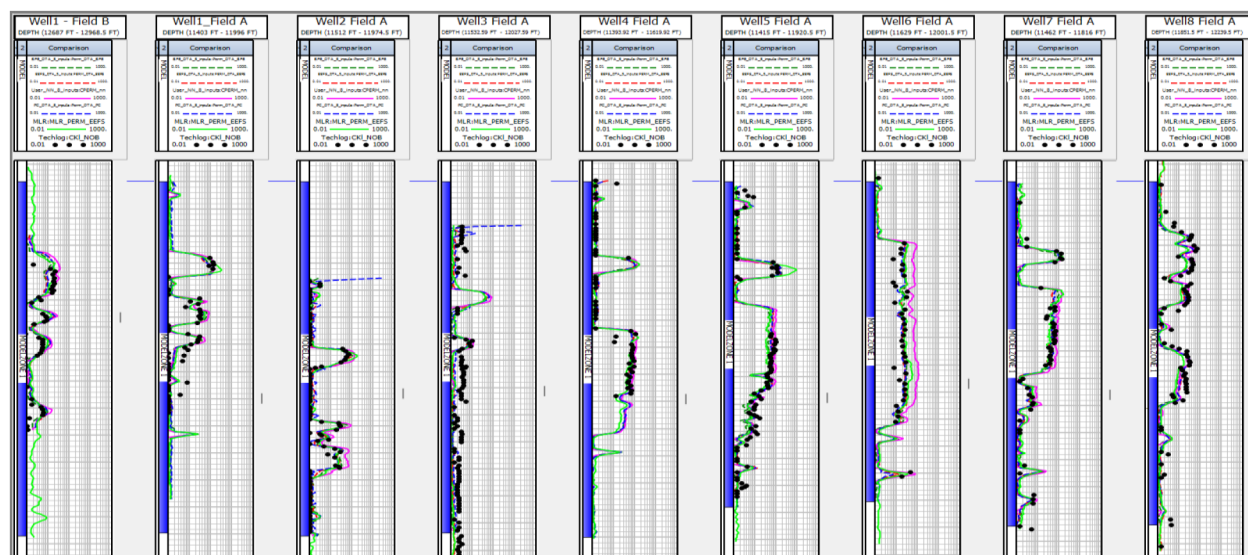


Figure 10—Permeability prediction – All cored wells kept blind to test the model's performance- Field A & B

In Field A and Field B, overall model performance was satisfactory in capturing the majority of predictive zones. However, specific wells exhibited notable deviations eg: Well 6 – Field A and Well 2 – Field A showed overestimation of permeability by the Neural Network (NN) model. This behavior suggests that NN may be sensitive to input variability and prone to extrapolation errors, especially in zones with limited core calibration. The discrepancy highlights the importance of optimal model selection and the need for robust feature selection strategies when applying ML models to permeability prediction. These observations reinforce the reliability of DTA-PC and DTA-EEFS, which demonstrated better alignment with core data across wells and maintained predictive stability in blind test scenarios.

Comparative plots were generated for all blind wells from Field C, which were excluded from the training dataset. These wells exhibit significant inter-well variability in permeability, providing a robust basis for evaluating model generalization and predictive performance under high uncertainty conditions.

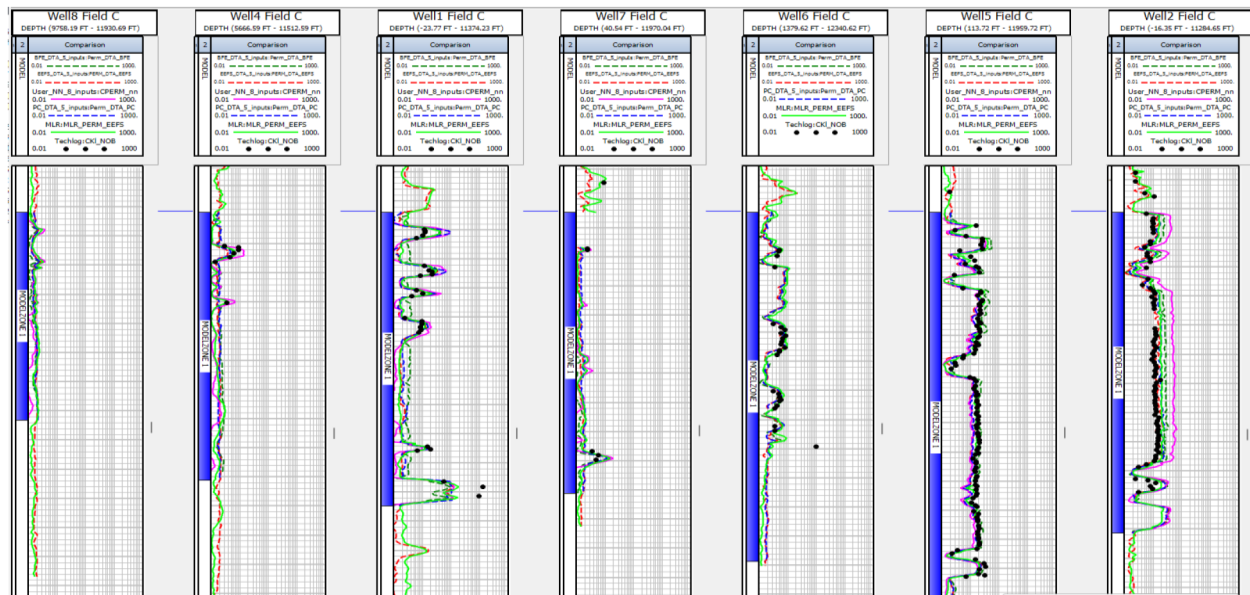


Figure 11—Permeability prediction – All cored wells kept blind to test the model's performance- Field C

Well 2 Field C well exhibits the highest variability in permeability prediction across models. Model Behavior on Neural Network (NN), DTA-BFE, and MLR models failed to capture the actual core permeability, showing significant scatter and deviation from measured values. DTA-EEFS and DTA-PC models, however, accurately captured the permeability trend, aligning closely with core data and demonstrating robustness in blind prediction scenarios. Well 1 – Field C model performance varies vertically across the well section. Upper Section Models tend to overestimate permeability, likely due to extrapolation beyond the training data range. Lower Section Models underperform, attributed to limited core data points and the presence of statistical outliers (as flagged by IQR-based filtering). Model Behavior Despite these challenges, DTA-PC and DTA-EEFS models outperformed others, showing better alignment with core permeability and maintaining predictive stability across the full depth range in Field C.

Uncertainty Analysis

Uncertainty quantification was a critical component of this study, aimed at evaluating the reliability and robustness of machine learning models used for permeability prediction.

Sensitivity plots and error propagation assessments were conducted to understand how input variability affects model outputs across different geological settings.

- Model Sensitivity:** DTA models demonstrated the lowest sensitivity to input fluctuations, maintaining consistent predictive performance across blind wells. In contrast, Neural Networks (NN) showed higher sensitivity, particularly in extrapolated zones with extreme permeability values, indicating reduced reliability without robust feature selection.
- Error Propagation:** Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics were used to quantify prediction uncertainty. DTA models consistently achieved lower RMSE (0.2–0.32) and MAE (0.09–0.15), while MLR models exhibited higher error ranges (RMSE: 0.55–0.81, MAE: 0.21–0.25), confirming their limited capability in handling complex reservoir heterogeneity.

- **Implications for Reservoir Risk Management:** The variability in model sensitivity and error metrics underscores the importance of integrating uncertainty analysis into permeability workflows. By identifying zones of high predictive uncertainty, operators can better manage reservoir risks and prioritize data acquisition or model refinement in critical areas.

Conclusion and Findings

This study presents a comprehensive framework for permeability prediction in complex carbonate reservoirs using machine learning (ML) techniques, core data calibration and uncertainty quantification.

The following key conclusions and findings were drawn

Core Data Integration: Incorporating Routine Core Analysis (RCA), Mercury Injection Capillary Pressure (MICP) and thin section petrography provided a strong calibration base, reducing bias and improving model alignment with actual reservoir conditions.

EDA techniques such as heatmaps, histograms, box plots, and cross plots were instrumental in identifying key relationships and data quality issues. Heatmaps revealed strong inter-variable correlations and guided feature selection, while histograms and box plots highlighted lithology-specific variability and outliers. These insights informed the grouping strategy by LITHO, SRT, and ZONE, which proved critical for building representative training datasets and improving model generalization.

Domain Transfer Analysis (DTA) consistently delivered the highest predictive accuracy, with R^2 values exceeding 0.88 in training and 0.79 in test datasets. It also achieved the lowest MAE and RMSE values, confirming its robustness and generalization capability across blind wells.

Neural Networks (NN) effectively captured nonlinear permeability trends but showed sensitivity to extrapolated zones, emphasizing the need for careful input validation.

Multiple Linear Regression (MLR) underperformed in terms of statistical metrics but provided visually acceptable cross plots, indicating limited utility in simpler scenarios.

The use of Experienced Eye Feature Selection (EEFS), Pearson's Correlation (PC), and Backward Feature Elimination (BFE) significantly improved model precision and reduced noise. EEFS leveraged domain knowledge to rank features based on physical relevance, enhancing the interpretability and reliability of the models.

Sensitivity analysis and error propagation studies demonstrated varying levels of model reliability, enabling better risk management in reservoir characterization. The workflow supports predictive confidence across different geological settings, aiding decision-making in exploration and production planning. The study successfully integrated EDA, feature selection, ML modeling and uncertainty analysis into a unified workflow.

Future work may focus on field-scale deployment, hybrid model development, and cloud-based implementation to further enhance scalability and operational efficiency.

Acknowledgments

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Nomenclature

BFE	Backward Feature Elimination
CPOR	Core Porosity
DTA	Domain Transfer Analysis
DTA-BFE	Domain Transfer Analysis- Backward Feature Elimination
DTA-EEFS	Domain Transfer Analysis- Experienced Eye Feature Selection

DTA-PC	Domain Transfer Analysis- Pearson's Correlation
EDA	Exploratory Data Analysis
EEFS	Experienced Eye Feature Selection
GR	Gamma Ray
IQR	Interquartile Range
KL	Core Permeability
KNN	K-Nearest Neighbors
LITHO	Lithology Grouping
MAE	Mean Absolute Error
MICP	Mercury Injection Capillary Pressure
ML	Machine Learning
MLR	Multiple Linear Regression
NN	Neural Network
NPHI	Neutron Porosity
PC	Pearson's Correlation
PEF	Photoelectric Factor
QC	Quality Check
RCA	Routine Core Analysis
RF	Random Forest
RHOB	Bulk Density
RMSE	Root Mean Square Error
RT	True Resistivity
RXO	Flushed Zone Resistivity
SRT	Static Rock Type
SVR	Support Vector Regression
TVDSS	True Vertical Depth Subsea
VDOL	Volume of Dolomite
VLIME	Volume of Limestone
ZONE	Geological Sequence Boundary Grouping
LPI	Large Pore Index

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Appendix - Exploration Data Analysis (EDA)

Univariate Analysis

Univariate analysis involves the statistical examination of a single variable at a time. In petrophysical workflows, this technique is essential for understanding the distribution, central tendency, and variability of key input parameters used to train variables that is not available in the data set eg: RHOB, NPHI, GR, Core porosity, Core Grain density, Core Permeability within the group framework.

By analyzing each variable independently, it becomes possible to identify data quality issues, detect outliers, and assess the representativeness of the dataset before applying multivariate or machine learning techniques.

Heatmaps

A heatmap of R^2 values was used to assess variable correlations with Core Permeability (CPerm) to identify key predictors for permeability within each lithology group (Figure A 1).

- LITHO 1: VDOL and VLIME showed strong inter-correlation but weak links to CPerm due to limited permeability range.
- LITHO 2: Core Porosity had a higher R^2 with Prem, while other variables showed low correlation.

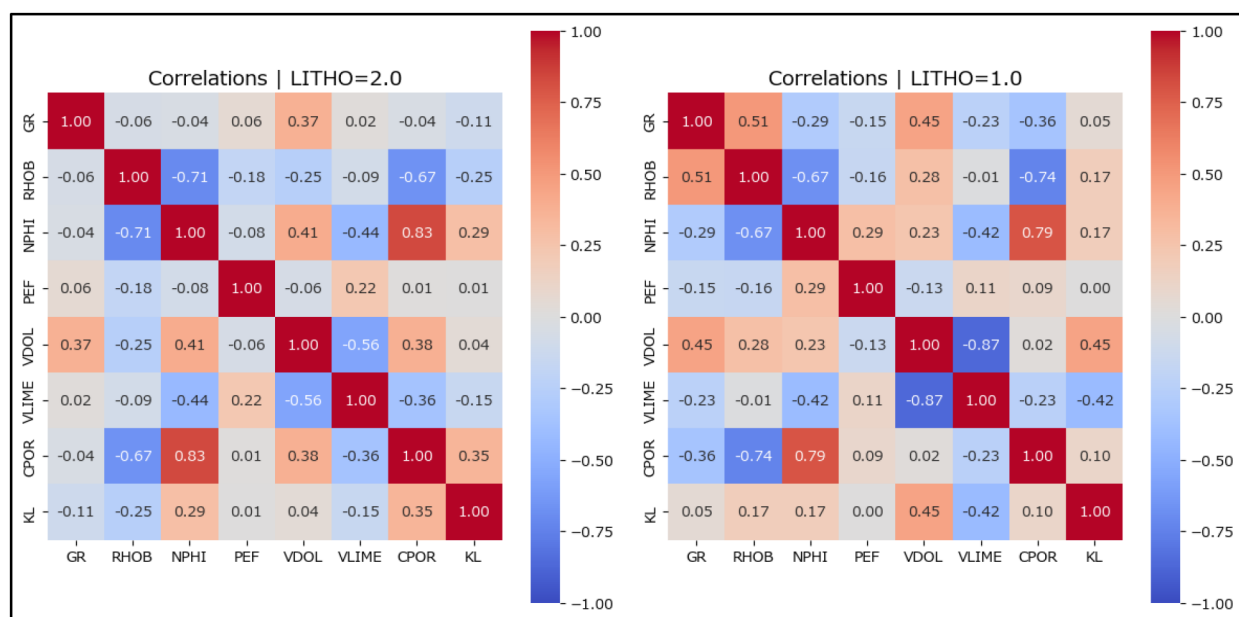


Figure A1—Pairwise correlation heatmaps Group by LITHO type

Histograms

Histograms were used to evaluate the distribution of log and core data across different lithologies.

- LITHO-01 exhibits a narrower distribution range, suggesting more uniform petrophysical properties within this group.
- In contrast, LITHO-02 shows a broader and more dispersed distribution, reflecting greater variability in the data.

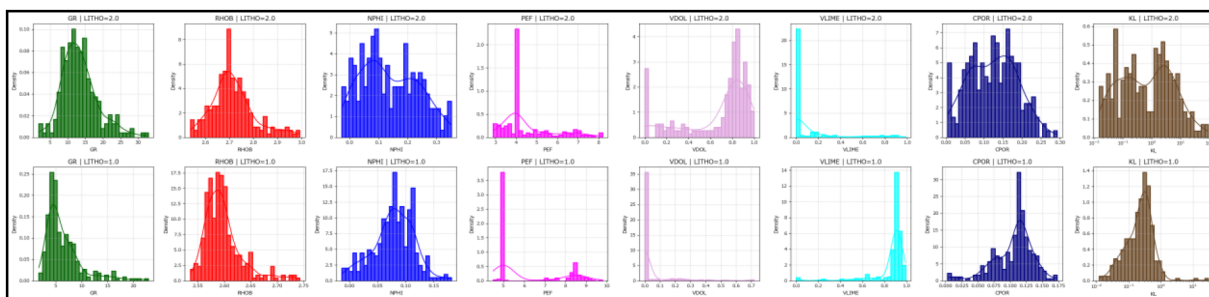


Figure A2—Histogram by LITHO-2 / LITHO-1: a) GR b) RHOB c) NPHI d) PEF e) VDOL f) VLIME g) CPOR h) KL

This distinction is critical for understanding heterogeneity and guiding lithology-specific grouping while performing multivariate and modeling exercise for permeability prediction.

Box Plots

Box plot analysis (Figure A 3, Figure A 4) was performed to evaluate the relationship between permeability and key input variables. The plots reveal distinct median values across variables, indicating potential predictive links with permeability. This variation suggests that certain inputs may serve as reliable indicators for permeability modeling, supporting their inclusion in data-driven reservoir characterization workflows.

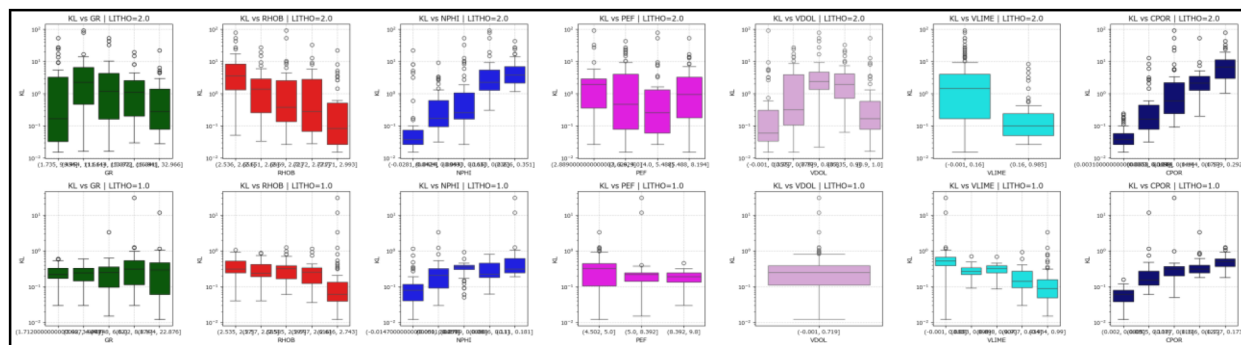


Figure A3—Box plots by LITHO-2 / LITHO-1 of KL vs. a) GR b) RHOB c) NPHI d) PEF e) VDOL f) VLIME g) CPOR

Box plots of KL Group by LITHO / SRT & ZONE:

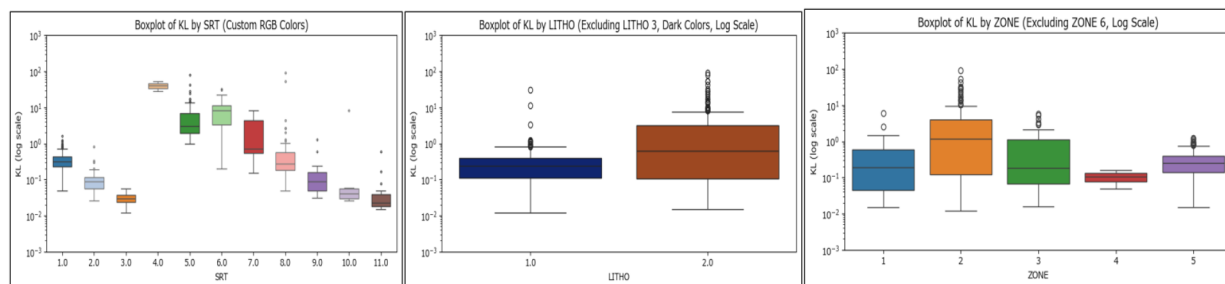


Figure A4—Box plots of KL vs. a) SRT b) LITHO c) ZONE

Bivariate / Multivariate Analysis under EDA

Log data based Multivariate Analysis was undertaken to establish data relationships prior to building training models (Figure A 5, Figure A 6).

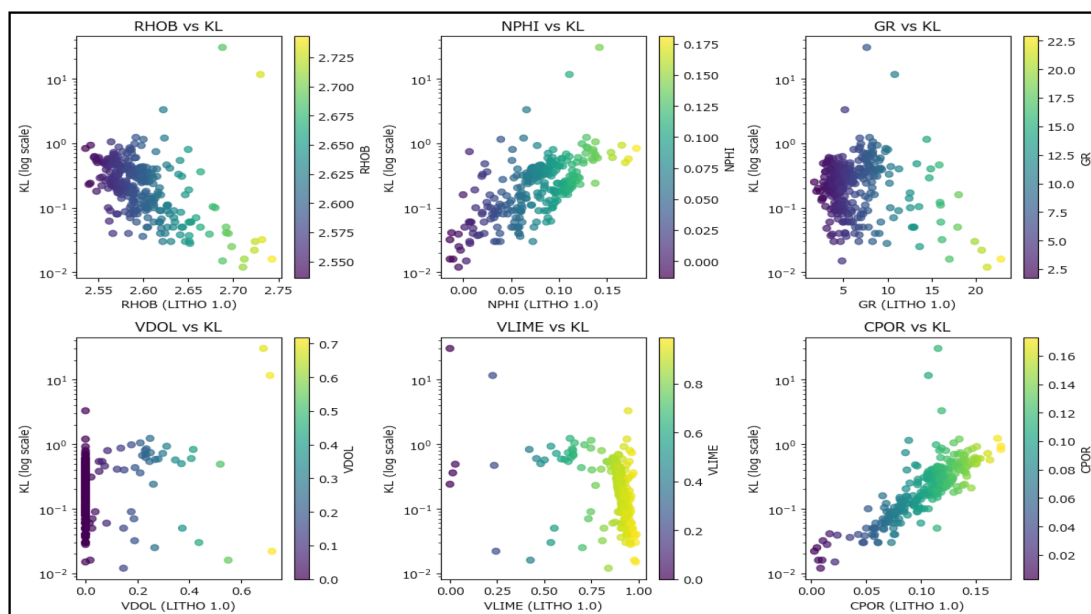


Figure A5—Scatter plot of Well log data vs. Permeability -KI (LITHO-01)

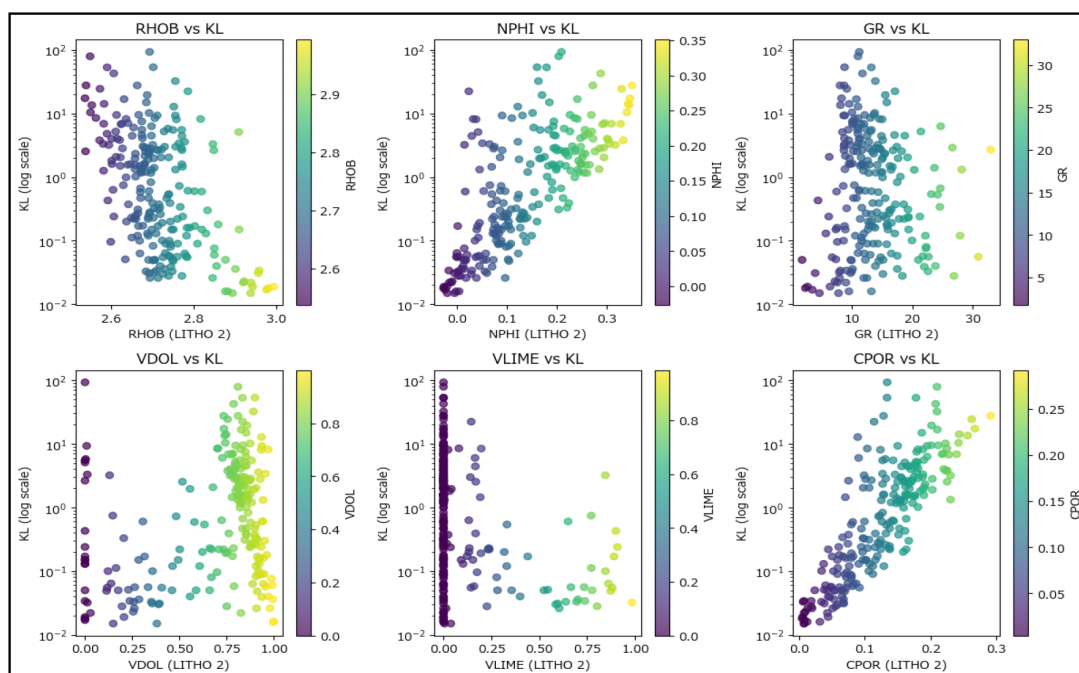


Figure A6—Scatter plot of Well log data vs. Permeability -KI (LITHO-02)

To improve model predictability, a ratio-based approach was applied to generate new variables from existing log and core data inputs. A total of 62 derived ratios and products were tested to evaluate their correlation with core permeability.

The analysis identified the top five AI-generated variables with the highest R^2 values, which are proposed for inclusion in the final dataset prior to model development. These engineered features are expected to enhance the robustness and accuracy of permeability prediction.

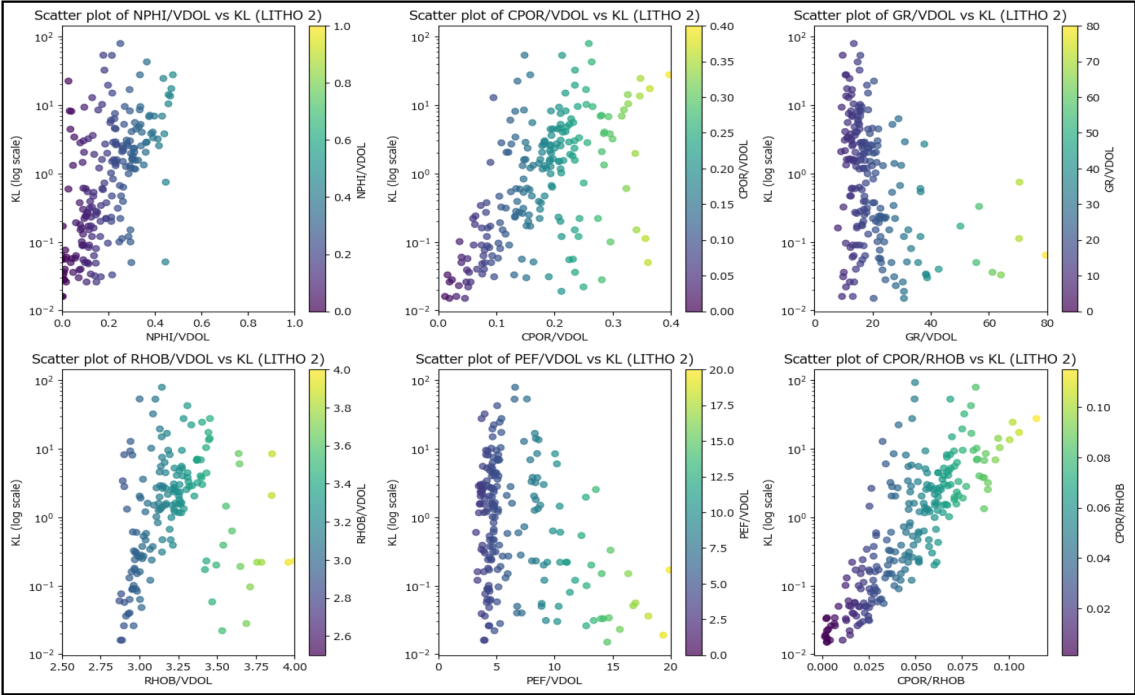


Figure A7—Scatter plot of AI Generative Additional Variables (LITHO-02)

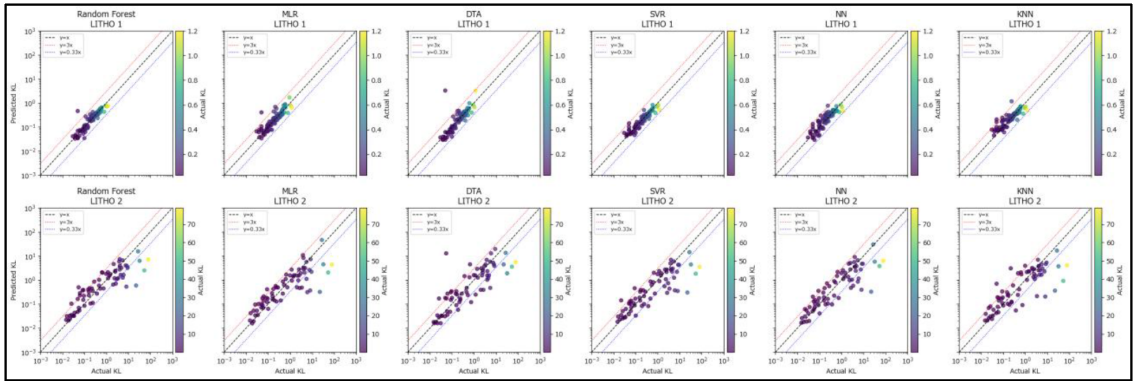


Figure A8—Model predictions for LITHO-01 & LITHO-02: Train 70% / Test 30% a) RF b) MLR c) DTA d) SVR e) NN f) KNN

Model Development to test variable statistical models:

Table A1—AI Generative Additional Variables

Feature	Type	Correlation
NPHI / VDOL	Ratio	0.511
CPOR / VDOL	Ratio	0.487
GR / VDOL	Ratio	0.443
RHOB / VDOL	Ratio	0.387
CPOR / VDOL	Ratio	0.370

Table A2—Summary of prediction statistics for all models (combined LITHO groups) without using hyper tuning parameters and feature selection

	Model	MSE	MAE	R2
1	DTA	0.22	0.29	0.59
2	KNN	0.16	0.25	0.69
3	MLR	0.14	0.23	0.74
4	NN	0.13	0.22	0.74
5	Random Forest (RF)	0.12	0.21	0.78
6	SVR	0.13	0.21	0.76